

On Variance Reduction in Learning Mean Flows

Juanwu Lu
Purdue University
juanwu@purdue.edu

Ziran Wang
Purdue University
ziran@purdue.edu

Abstract

Mean Flows enable distillation-free one-step generation by learning two-parameter average-velocity fields from noise to data distributions, but their training is notorious for non-decreasing loss and high gradient variance. We establish a theoretical framework that explains the phenomenon through the lens of *variance reduction with a control variate*. We identify two distinct statistical roles for the conditional velocity v_{cond} in training: as an unbiased regression target and as the Monte Carlo control variate for the tangent in the Jacobi-vector-product (JVP), but with a statistically suboptimal coefficient. The latter inflates per-step gradient variance and produces the non-decreasing loss. We derive the optimal tangent-mixing coefficient β^* in closed form and show that a family of concurrent works correspond to different practical realizations of the same optimum. To validate the framework, we instantiate the $\beta = 1$ corner via an EMA-derived proxy with a flow-matching anchor and run a controlled β -sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at both toy and DiT-B/4 / ImageNet-256 scale. The sweep recovers the predicted bias-variance ordering at every matched-step checkpoint, with $3\times \sim 5\times$ gradient-variance reduction and up to 54% SW_1 improvement on 2-D benchmarks, and $\approx 5\times$ reduction in gradient noise ratio at DiT scale. The same DiT-scale measurement also exposes a quantitative *FID-vs-MSE landscape mismatch*: the FID-axis bias amplification across $\beta = 0.5/\beta = 1$ ($\sim 1.4:1$) is sublinear in the gradient-MSE bias coefficient (1:4), so the FID-optimal β shifts beyond the gradient-MSE interior minimizer to the unbiased corner $\beta = 0$.

1 Introduction

Deep generative models that leverage neural networks to parametrize and sample from unknown data distributions have achieved remarkable success [1, 2, 3, 4, 5]. Among these models, flow-based generative models [6, 7, 8, 9, 10, 11, 12] learn a continuous-time transformation between a simple prior and the data distribution. Conditional flow matching [10, 11] and stochastic interpolants [12, 13] further enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [1, 14, 2]. However, integration at inference time remains a challenge due to its computational costs and numerical instability, motivating works on *one-step generation*, including progressive distillation [15], distribution matching [16, 17], consistency models [18, 19], and flow map matching [20]. Nevertheless, most of them require a pre-trained distillation teacher.

Mean Flows [21] provide a *distillation-free* alternative by learning a two-parameter average velocity field $u(x, r, t)$ through regressing a total-derivative identity between the average and instantaneous velocity. In principle, this identity yields exact self-supervision. In practice, however, training is plagued by *non-decreasing loss* and prohibitively *high gradient variance* [22, 23]. A plethora of concurrent works have proposed remedies that target different mechanisms. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α -curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a learned marginal-velocity head. Re-MeanFlow [24] preprocesses the data with rectified-flow straightening to reduce trajectory curvature. Terminal Velocity Matching [25] sidesteps the spatial Jacobian by differentiating with respect to the terminal time. Functional Mean Flows [26] extend MeanFlow to Hilbert spaces and study marginal-conditional consistency for functional data. Each fix is empirically effective in isolation, but there is no theory that explains *why* the original objective is unstable or what these fixes have in common.

In this paper, we establish our theory around a single research question:

Why do stochastic tangents destabilize learning Mean Flows?

We answer this question through the lens of *variance reduction*. Specifically, we identify that the conditional velocity $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$ plays two statistically distinct roles in the MeanFlow loss. As the regression target, it is an *unbiased* estimator of the marginal velocity $\mathbf{v}$. As the tangent inside the Jacobian-vector product (JVP) of the total-derivative identity, however, the same $\mathbf{v}_{\text{cond}}$ acts as a Monte Carlo *control variate* for the inaccessible marginal velocity, with a statistically suboptimal coefficient. The bias-variance trade-off generated by this substitution governs gradient variance through the quadratic form $\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}} \mathbf{J}^\top \mathbf{g}$, where $\mathbf{g} = \nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}}$ and $\mathbf{J}$ is a spatial-Jacobi factor. The conditional fluctuation $\mathbf{v}_{\text{cond}} - \mathbf{v}$ is amplified by $\mathbf{J}$ at each gradient step, and stop-gradient prevents this amplification from reaching the optimizer, leading to the empirical pathology.

Following the theory, we derive the optimal tangent control-variate coefficient in closed form and show that it converges to a deterministic-tangent estimator in the practical regime where a marginal-velocity estimator is available. This result *explains* why deterministic tangents (e.g., *EMA teachers and learned velocity heads*) all stabilize training: they are different practical instantiations of the same statistical optimum. To validate the framework empirically, we instantiate the $\beta = 1$ corner of β^* via an EMA-derived proxy with a flow-matching anchor (one extra JVP per step relative to vanilla MeanFlow), and run a controlled β -sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at toy and DiT-B/4 / ImageNet-256 scale. Direct measurement of per-step gradient variance recovers the predicted non-decreasing loss component $\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}} \mathbf{J}^\top)$ and a $3\times \sim 5\times$ reduction at $\beta = 1$ across 2-D benchmarks. At DiT scale the same sweep also exposes a quantitative *FID-vs-MSE landscape mismatch*: the FID ordering matches the bias-variance prediction across all four β values, but the FID-axis bias amplification is sublinear in the MSE-axis coefficient—so the FID-optimal β shifts beyond the gradient-MSE interior minimizer to the unbiased corner. Our contributions in this paper are as follows:

- C1** We prove that per-step gradient variance in the original MeanFlow scales unboundedly with $\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}} \mathbf{J}^\top \mathbf{g}$ and the stop-gradient operator induces a *semi-gradient gap* that hides this term from the optimizer.
- C2** We frame the JVP tangent as a control variate for the marginal velocity, derive the optimal coefficient β^* in closed form, and unify concurrent works by showing that they each correspond to a different practical realization of β^* .
- C3** We empirically validate the framework via a controlled β -sweep at both 2-D and DiT-B/4 / ImageNet-256 scale, instantiating the $\beta = 1$ corner with an EMA proxy and flow-matching anchor. We observe $3\times \sim 5\times$ direct gradient-variance reduction, up to 54% SW_1 improvements on toy benchmarks, the predicted bias-variance FID ordering at every matched-step DiT checkpoint, and a quantitative FID-vs-MSE landscape mismatch (1.4:1 FID-axis bias amplification across $\beta = 0.5/\beta = 1$ vs 1:4 on the MSE axis).

2 Preliminaries

Flow-Matching. Let $\mathbf{x}$ denote a data sample in $\mathcal{X} \subset \mathbb{R}^d$. Flow-matching generative models [10, 11] learn a time-dependent diffeomorphism $\psi(\mathbf{x}, t) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$ by parameterizing a velocity field $\mathbf{v}(\mathbf{x}, t) \triangleq \frac{d}{dt} \psi(\mathbf{x}, t)$. Since the exact *marginal velocity field* $\mathbf{v}(\mathbf{x}, t)$ is generally inaccessible, conditional flow matching instead learns a conditional field $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0) = \mathbf{x}_1 - \mathbf{x}_0$ with $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$. The following lemma justifies the substitution. See Appendix B for its proof.

Lemma 1. *Given vector fields $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)$ generating probability paths $p(\mathbf{x}, t \mid \mathbf{x}_0)$, for any $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the expected conditional velocity field is equal to the marginal velocity field*

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0), t \mid \mathbf{x}} [\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)]. \quad (1)$$

One-step Mean Flows. Iterative integration of $\mathbf{v}(\mathbf{x}, t)$ at inference is expensive and can be numerically unstable. To bypass it, Mean Flows [21] learn a two-parameter average velocity field $\mathbf{u}(\mathbf{x}, r, t)$ through regressing the identity

$$(t - r) \mathbf{u}(\mathbf{x}, r, t) = \int_r^t \mathbf{v}(\mathbf{x}, \tau) d\tau. \quad (2)$$

Differentiating with respect to the terminal time t yields

$$\mathbf{u}(\mathbf{x}, r, t) + (t - r) \frac{d}{dt} \mathbf{u}(\mathbf{x}, r, t) = \mathbf{v}(\mathbf{x}, t), \quad \frac{d}{dt} \mathbf{u} = \partial_{\mathbf{x}} \mathbf{u} \cdot \mathbf{v}(\mathbf{x}, t) + \partial_t \mathbf{u}, \quad (3)$$

where the total derivative $\frac{d}{dt} \mathbf{u}$ is essentially a JVP, denoted by $\text{JVP}(\mathbf{u}, (\mathbf{x}, r, t), (\mathbf{v}(\mathbf{x}, t), 0, 1))$. The original MeanFlow paper [21] replaces *both* occurrences of the marginal field with the conditional field $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$, giving the MeanFlow loss

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r, t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\text{JVP}(\mathbf{u}_{\boldsymbol{\theta}}, (\mathbf{z}, r, t), (\mathbf{v}_{\text{cond}}, 0, 1))] - \mathbf{v}_{\text{cond}}\|_2^2 \right], \quad (4)$$

with $r, t \sim \mathcal{U}(0, 1)$, $r \leq t$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I}_d)$, $\mathbf{z} = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$, and the stop-gradient $\text{sg}[\cdot]$ avoiding double backpropagation through the JVP.

3 Variance Reduction in Learning Mean Flows

The MeanFlow loss in equation 4 is empirically known to be *non-decreasing* and to suffer from *high-variance gradients* [22, 23]. We investigate the statistical mechanism behind this pathology. Importantly, our analysis isolates the role of the conditional velocity field $\mathbf{v}_{\text{cond}}$ when it appears as the JVP tangent, and identifies that it acts as a Monte Carlo control variate for the inaccessible marginal velocity, with a coefficient that the original MeanFlow fixes to a suboptimal value. We derive the optimal coefficient and establish a connection to practical fixes in concurrent works.

3.1 Two roles of the conditional velocity field

With Reynolds decomposition, we express the conditional velocity by $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{v}'$ with $\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ according to equation 1 and $\Sigma_{\mathbf{v}'} \triangleq \text{Cov}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}']$. The MeanFlow loss in equation 4 substitutes $\mathbf{v}_{\text{cond}}$ in two distinct positions: as the *regression target* $-\mathbf{v}_{\text{cond}}$, and as the *tangent* inside the total derivative. Carrying the decomposition through per-sample loss $\ell_{\text{MF}}(\boldsymbol{\theta})$ yields:

$$\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\ell_{\text{MF}}(\boldsymbol{\theta})] = \|\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}\|_2^2 + \text{sg}[\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})], \quad (5)$$

where $\mathbf{J} \triangleq (t - r) \partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} - \mathbf{I}_d \in \mathbb{R}^{d \times d}$ is a *Jacobi factor* determined by the Jacobian matrix $\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}}$ and $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}} \triangleq \mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} \mathbf{v} + \partial_t \mathbf{u}_{\boldsymbol{\theta}}] - \mathbf{v}$ is the deterministic mean-field residual. Since the residual $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the exact loss we aim to minimize given by equation 3, the trace term leads to the non-decreasing loss and eliminates *iff* the Jacobi satisfies $\partial_{\mathbf{x}} \mathbf{u}_{\boldsymbol{\theta}} = (t - r) \mathbf{I}_d$. As a result, we identify two roles of the conditional velocity field in this decomposition:

Unbiased Target. As the regression target, $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *linearly* into the residual, contributing the *irreducible* per-sample noise floor $\text{Tr}(\Sigma_{\mathbf{v}'})$. This contribution is independent of $\boldsymbol{\theta}$ and reflects the natural Monte Carlo cost of replacing the inaccessible $\mathbf{v}$ by a per-sample proxy.

Variance Amplifier. As the JVP tangent, the same $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *multiplicatively* through the Jacobi factor $\mathbf{J}$, contributing the *amplified* term $\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})$. It scales unboundedly with the spectral magnitude of $\mathbf{J}$. We identify this amplification as the dominant driver of variance at the gradient level:

Theorem 2 (Jacobian Variance Amplification). *Let $\mathbf{g} \triangleq \nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}}(\mathbf{x}_t, r, t) \in \mathbb{R}^{d \times p}$ be the gradient of the average velocity. The conditional per-step gradient variance of $\ell_{\text{MF}}(\boldsymbol{\theta})$ in equation 4 satisfies*

$$\text{Var}[\nabla_{\boldsymbol{\theta}} \ell_{\text{MF}} | \mathbf{x}_t] \propto \mathbf{g}^{\top} \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top} \mathbf{g}.$$

See the proof in Appendix D. Our Theorem 2 identifies that the gradient variance grows with $\|\mathbf{J}\|^2$, saturates only at the irreducible noise floor when $\mathbf{J} \rightarrow \mathbf{0}$, and is *uncontrolled* by any term in the loss due to the use of the stop-gradient in the original MeanFlow. We notice that, without the stop-gradient operator, the optimizer could in principle reduce variance by driving $\mathbf{J} \rightarrow \mathbf{0}$. This establishes a *semi-gradient gap*, formally,

Theorem 3 (Semi-Gradient Gap). *The gradient of the MeanFlow loss $\mathcal{L}_{\text{MF}}$ with and without the stop-gradient operator differ by*

$$\underbrace{2(t - r) \mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\boldsymbol{\theta}}) \cdot \mathbf{r}_{\boldsymbol{\theta}}]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E}[\nabla_{\boldsymbol{\theta}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})]}_{\text{variance-driven gradient difference}}.$$

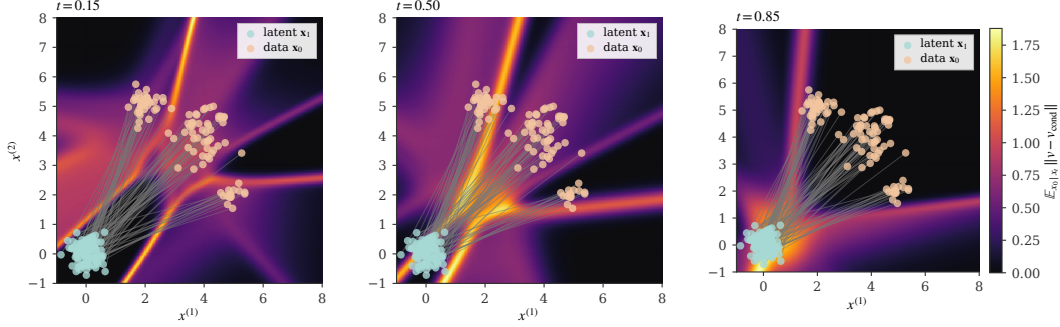


Figure 1: Conditional-marginal velocity gap with p_{data} a 2-D Gaussian mixture with three components. The spatial heatmap of $\mathbb{E}_{x_0|x_t} \|\mathbf{v}(\mathbf{x}_t, t) - \mathbf{v}(\mathbf{x}_t | \mathbf{x}_0)\|$ shows that the gap is heterogeneous and concentrates in mode-mixing regions where conditional paths overlap.

At convergence, the mean-field difference vanishes ($\mathbf{r}_\theta \rightarrow \mathbf{0}$) and the variance-driven term dominates. Since the optimizer is *blind* to the variance-amplification term in the original MeanFlow, the term is silently accumulating, producing the non-decreasing-loss empirical pathology. Figure 1 visualizes the an example case. We showcase that $\|\mathbf{v}'\|$ is non-zero, increases toward the latent, and concentrates spatially in mode-mixing regions, contributing to the non-decreasing loss.

3.2 Rethinking tangent as a control variate

The amplification in Theorem 2 arises due to using $\mathbf{v}_{\text{cond}} = \mathbf{v} + \mathbf{v}'$ as the JVP tangent injects the zero-mean fluctuation $\mathbf{v}'$. This actually corresponds to the canonical structure of a Monte Carlo *control variate* [27], where $\mathbf{v}_{\text{cond}}$ is a stochastic estimator of the inaccessible $\mathbf{v}$ with known mean, and the coefficient on the noise term $\mathbf{v}'$ can be a free parameter that determines the bias-variance trade-off. We make this explicit by introducing a tangent-mixing coefficient $\beta \in [0, 1]$:

$$\tilde{\mathbf{v}}_{\text{tang}}(\beta) \triangleq (1 - \beta) \mathbf{v}_{\text{cond}} + \beta \hat{\mathbf{v}} = \mathbf{v} + (1 - \beta) \mathbf{v}' + \beta \mathbf{b}, \quad (6)$$

where $\hat{\mathbf{v}}$ can be any deterministic proxy for the marginal velocity (e.g., a *velocity network*) with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}$ that is deterministic given $\mathbf{x}_t$. Herein, $\beta = 0$ recovers vanilla MeanFlow and $\beta = 1$ replaces the tangent entirely with the deterministic proxy.

Bias-variance trade-off. Substituting $\tilde{\mathbf{v}}_{\text{tang}}(\beta)$ for $\mathbf{v}_{\text{cond}}$ in the JVP tangent and propagating through the loss, the per-sample gradient writes

$$\mathbf{g}^{(\beta)} = 2 \mathbf{g}^\top \left[\mathbf{r}_\theta + ((1 - \beta) \mathbf{J} - \beta \mathbf{I}) \mathbf{v}' + \beta (\mathbf{J} + \mathbf{I}) \mathbf{b} \right], \quad (7)$$

where the noise term scales the conditional fluctuation $\mathbf{v}'$ by the matrix $((1 - \beta) \mathbf{J} - \beta \mathbf{I})$, and the bias term scales the proxy error $\mathbf{b}$ by $\beta (\mathbf{J} + \mathbf{I})$. This establishes the following theorem.

Theorem 4 (Optimal control-variate coefficient). *Define the conditional MSE of the per-sample gradient equation 7 as its mean-squared error against its noise- and bias-free target $2 \mathbf{g}^\top \mathbf{r}_\theta$, taken over the randomness of $\mathbf{v}'$ at fixed $\mathbf{x}_t$:*

$$M(\beta) \triangleq \mathbb{E}_{\mathbf{v}'|x_t} [\|\mathbf{g}^{(\beta)} - 2 \mathbf{g}^\top \mathbf{r}_\theta\|_2^2].$$

Under the scalar-isotropic approximation $\Sigma_{\mathbf{v}'} \approx \sigma^2 \mathbf{I}_d$ and $\mathbf{J} \approx \kappa \mathbf{I}_d$, $M(\beta) = \beta^2(\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1 - \beta)\kappa - \beta)^2$ and admits a unique minimizer

$$\beta^* = \underbrace{\frac{\kappa}{\kappa + 1}}_{\text{noise-cancellation}} \cdot \underbrace{\frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}}_{\text{shrinkage}},$$

with optimum value $M(\beta^) = \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. For all $\kappa, \|\mathbf{b}\|^2 > 0$, $M(\beta^*)$ strictly dominates both corner MSEs $M(0) = \sigma^2 \kappa^2 d$ (vanilla MeanFlow) and $M(1) = (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d$ (deterministic-tangent estimator with bias).*

Table 1: Practical fixes in concurrent works for training pathology in MeanFlow as coefficient in control variate. Each method either drives $\beta \rightarrow 1$ with a deterministic proxy, or reduces the amplification factor $\|\mathbf{J}\|^2$ directly.

Method	Mechanism in our framework	Practical realization
AlphaFlow [22]	avoid high- κ regimes	curriculum schedule
Improved MF [23]	deterministic tangent ($\beta \rightarrow 1$)	separate velocity head + sg
TVM [25]	remove the spatial Jacobian ($\mathbf{J} \rightarrow \mathbf{0}$)	terminal-time differentiation
Re-MeanFlow [24]	reduce $\ \mathbf{J}\ $	rectified-flow preprocessing
VaMF (ours)	deterministic tangent ($\beta \rightarrow 1$)	EMA proxy + FM anchor

The proof is a direct application of bias-variance decomposition to equation 7 and is given in Appendix D. The two factors of β^* have crisp interpretations: $\kappa/(\kappa+1)$ is the coefficient that *cancels the Jacobian-amplified noise* via destructive interference between the $\mathbf{J}\mathbf{v}'$ and $\mathbf{I}\mathbf{v}'$ contributions in equation 7; $\sigma^2 d/(\sigma^2 d + \|\mathbf{b}\|^2)$ is the standard James-Stein shrinkage that pulls β^* toward the unbiased corner when the proxy bias $\|\mathbf{b}\|$ is large. As the proxy bias shrinks ($\|\mathbf{b}\|^2 \rightarrow 0$), $\beta^* \rightarrow \kappa/(\kappa+1)$, and as the Jacobian magnitude grows ($\kappa \rightarrow \infty$), $\beta^* \rightarrow 1$.

Vanilla MeanFlow is generally suboptimal. Vanilla MeanFlow uses $\beta=0$ which is optimal only when $\kappa=0$ (i.e., $\partial_{\mathbf{z}}\mathbf{u}_{\theta} = \frac{1}{t-r}\mathbf{I}_d$) or when no deterministic proxy is available ($\|\mathbf{b}\| = \infty$). Both conditions fail in practice: trained backbones produce $\partial_{\mathbf{z}}\mathbf{u}_{\theta}$ matrices that are spectrally far from $\frac{1}{t-r}\mathbf{I}_d$, and an EMA copy of $\mathbf{u}_{\theta}$ provides a viable deterministic proxy at near-zero compute cost.

Concurrent fixes through this lens. Several remedies for MeanFlow’s training pathology in concurrent work can now be located in the same one-parameter family. Each fix corresponds to a different practical realization of the optimum, distinguished by *which proxy* they use for $\mathbf{v}$ and *which value of β* the construction implicitly selects:

- **Deterministic-tangent methods** (Improved MF [23], EMA tangents) take $\beta \rightarrow 1$ with a learned or EMA-derived proxy $\hat{\mathbf{v}}$. The amplified noise term is eliminated; the bias term $(\mathbf{J}+\mathbf{I})\mathbf{b}$ is controlled by training the proxy.
- **Reparameterization methods** (TVM [25]) sidestep the spatial Jacobian by reparameterizing the total derivative, effectively setting $\mathbf{J}=\mathbf{0}$ in our framework.
- **Curvature-reduction methods** (Re-MeanFlow [24]) reduce $\|\mathbf{J}\|$ via reflow preprocessing, lowering κ and the amplified-noise term.
- **Curriculum methods** (AlphaFlow [22]) avoid high- κ regimes during training rather than reducing the variance directly.

None of these methods explicitly identify the control-variate structure or derive β^* . We do, and we use the result both as an analytical lens for these methods and as the foundation for our practical training framework.

3.3 Empirical instantiation: Variance-Aware MeanFlow (VaMF)

To validate the framework empirically, we need a concrete realization of β^* that we can sweep at scale. We construct *Variance-Aware MeanFlow* (VaMF) as a controlled instance of the $\beta=1$ corner: the JVP tangent is replaced by an EMA-derived deterministic proxy $\hat{\mathbf{v}} = \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)$, and a small flow-matching anchor at $r=t$ keeps the proxy bias $\mathbf{b} = \hat{\mathbf{v}} - \mathbf{v}$ small so the bias term $\beta(\mathbf{J}+\mathbf{I})\mathbf{b}$ in equation 7 stays controlled. The regression target remains $\mathbf{v}_{\text{cond}}$ unchanged—only the tangent is replaced, exploiting the role asymmetry of Section 3.1. The construction adds one extra JVP per step relative to vanilla MeanFlow. Appendix C gives the full loss, training algorithm, and three propositions characterizing the resulting gradient form, residual bias, and the necessity of the target-tangent role split. In Section 5 we sweep $\beta \in \{0, 0.25, 0.5, 1\}$ with VaMF realizing the $\beta=1$ corner, the linear interpolation realizing $\beta \in \{0.25, 0.5\}$, and vanilla MeanFlow realizing $\beta=0$.

4 Related Work

MeanFlow and its training pathology. Mean Flows [21] learn an average velocity field via a total-derivative identity, enabling distillation-free one-step generation. The empirical pathology—non-decreasing loss with high gradient variance—was diagnosed in concurrent work from several angles. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α -curriculum to manage the tradeoff. Improved MeanFlow [23] observes that the original compound predictor improperly depends on the conditional velocity input and replaces it with a separate marginal-velocity head trained jointly with stop-gradient. Re-MeanFlow [24] addresses trajectory curvature through a reflow pre-processing step that straightens the conditional paths. Terminal Velocity Matching (TVM) [25] sidesteps the spatial Jacobian entirely by differentiating with respect to the terminal time. Functional Mean Flows [26] extends MeanFlow to Hilbert spaces and studies marginal/conditional consistency for functional data. Each of these works proposes an effective practical fix; in contrast, our work isolates the statistical role of the JVP tangent itself, identifies the conditional fluctuation v' as a control variate with the wrong coefficient inside a semi-gradient JVP, and derives the optimal coefficient. Concurrent fixes correspond to different practical realizations of this optimum (Table 1).

Variance reduction in flow training. Several recent works address variance in flow-based training from a complementary angle. Stable Velocity [28] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [29] reduces variance through temporal pair coupling; and preconditioned flow matching [30] addresses ill-conditioning via anisotropic preconditioning. Bertrand et al. [31] argue that stochastic-target variance is negligible for standard flow matching—consistent with our analysis, since the problematic amplification we identify is specific to the MeanFlow JVP, not the flow-matching loss itself. Rectified flows [32, 33] reduce trajectory curvature by iterative straightening, which by Theorem 2 reduces $\|J\|^2$ and hence the amplified noise term.

Control variates and target networks. The control-variate framework we apply has classical roots in Monte Carlo estimation [34]; it is also the analytical structure underlying target networks in deep RL [35] and self-distillation in self-supervised learning [36]. We make the connection explicit by interpreting the EMA-derived tangent in MeanFlow as a target-network-style proxy that drives the control-variate coefficient toward its optimum.

5 Experiments

We test three claims directly: (i) per-step gradient variance scales with $g^\top J \Sigma_{v'} J^\top g$ as predicted by Theorem 2; (ii) approaching the control-variate optimum $\beta \rightarrow 1$ via VaMF reduces gradient variance by a measurable factor; (iii) the variance reduction translates to better sample quality. All toy and DGMM experiments use a 3-layer MLP with 128 hidden units, 200,000 optimization steps, and three random seeds $\{42, 0, 1\}$. Code, configurations, and result JSONs are released with the paper.

5.1 Direct Measurement of Gradient Variance

Figure 2 reports the mean per-step gradient covariance trace $\text{Tr}(\text{Cov}[g])$ as a function of the tangent coefficient $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$ on four 2-D toy datasets (eight Gaussians, two moons, swiss roll, pinwheel). At each measurement step, we draw $K = 8$ independent mini-batches and compute the sample variance of their gradients across parameters. The variance is monotonically decreasing in β on every dataset, and the reduction at $\beta = 1$ relative to vanilla MeanFlow ($\beta = 0$) is $4.7\times$ on eight Gaussians, $3.8\times$ on two moons, $3.2\times$ on pinwheel, and $1.7\times$ on swiss roll. This directly verifies Theorem 2: replacing the conditional fluctuation v' in the JVP tangent with a deterministic proxy eliminates the dominant Jacobian-amplified contribution to gradient variance. The right panel further shows that the reduction holds across the full training trajectory, not just at convergence, confirming that the amplification is operative throughout training.

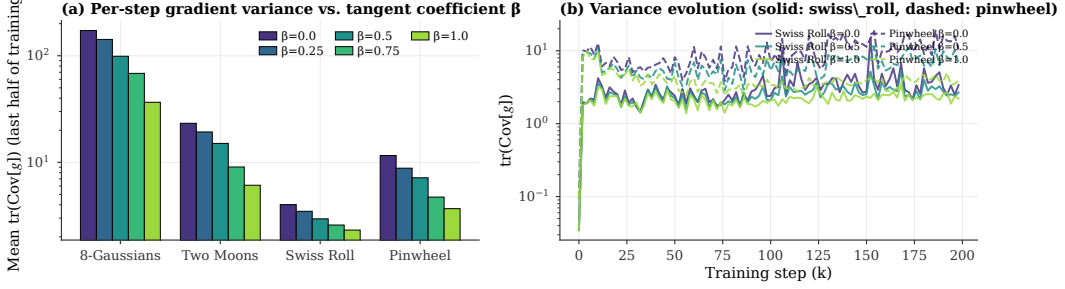


Figure 2: Direct gradient-variance measurement during training (Theorem 2 validation). (a) Mean $\text{Tr}(\text{Cov}[g])$ over the second half of training, per dataset, as a function of the tangent coefficient β . The variance drops monotonically with β , with $3.2\times \sim 4.7\times$ reduction at $\beta = 1$ on three of four datasets. (b) Variance trajectory across training on `swiss_roll` (solid) and `pinwheel` (dashed), confirming the reduction is operative throughout training, not just at convergence.

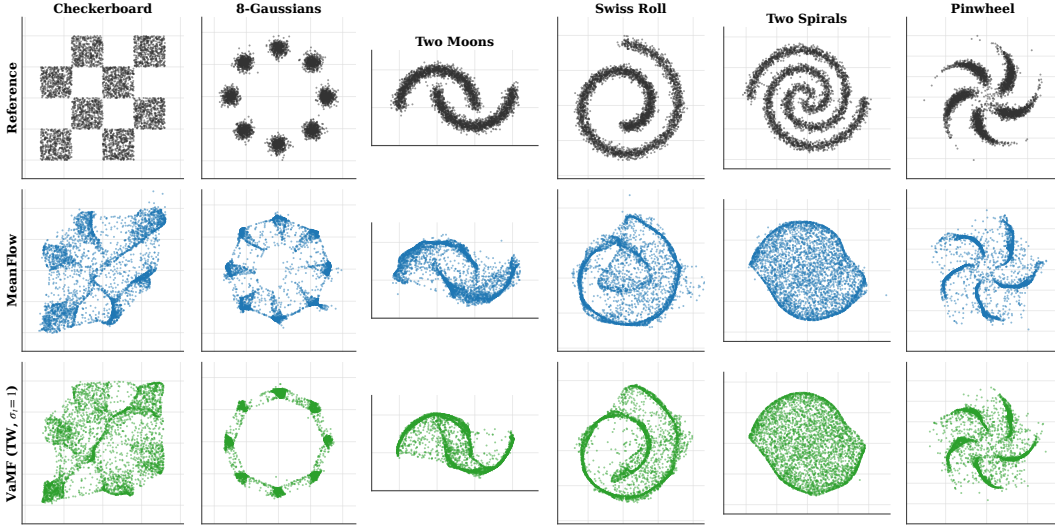


Figure 3: Reference distribution and final-step samples on the 2-D toy datasets at seed 42. VaMF (bottom row) recovers structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out, especially on `checkerboard` and `swiss_roll`.

5.2 Toy Benchmark

Table 2 reports SW_1 and SW_2 on six 2-D toy datasets averaged over three random seeds. VaMF (with EMA proxy + FM anchor, corresponding to $\beta \rightarrow 1$ in our framework) outperforms vanilla MeanFlow on five of six datasets, with relative SW_1 improvements of -23.7% (`checkerboard`), -13.2% (`eight_gaussians`), -22.5% (`two_moons`), -53.7% (`swiss_roll`), and -37.2% (`pinwheel`). On `two_spirals`, where MeanFlow already achieves the lowest absolute SW_1 (0.034) of any dataset, VaMF is marginally worse ($+7.9\%$ on a small absolute base). The SW_2 improvements broadly follow the SW_1 pattern, with even larger margins on `swiss_roll` (-66.0%) and `two_moons` (-47.6%) where mode-mixing creates heavy tails in the residual distribution. Figure 3 visualizes samples: VaMF recovers structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out.

5.3 High-Dimensional Scaling

We sweep the data dimensionality on a dense Gaussian mixture (DGMM)—64 overlapping isotropic components on a unit sphere of radius 1.5 in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$. Table 3 reports SW_1 at convergence (mean $\pm$ standard error over three seeds). MeanFlow and VaMF are within one standard error at every d , with VaMF edging out at $d \in \{2, 8, 16\}$ and MeanFlow tying or marginally winning

Table 2: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates.

Dataset	Metric	MeanFlow	VaMF
checkerboard	SW_1	0.175 ± 0.036	0.134 ± 0.029
	SW_2	0.234 ± 0.042	0.175 ± 0.046
eight_gaussians	SW_1	0.098 ± 0.016	0.085 ± 0.010
	SW_2	0.151 ± 0.022	0.146 ± 0.023
two_moons	SW_1	0.093 ± 0.028	0.072 ± 0.019
	SW_2	0.177 ± 0.068	0.093 ± 0.025
swiss_roll	SW_1	0.098 ± 0.058	0.045 ± 0.003
	SW_2	0.177 ± 0.151	0.060 ± 0.007
two_spirals	SW_1	0.034 ± 0.001	0.037 ± 0.001
	SW_2	0.044 ± 0.001	0.046 ± 0.001
pinwheel	SW_1	0.094 ± 0.011	0.059 ± 0.005
	SW_2	0.128 ± 0.014	0.076 ± 0.005

Table 3: DGMM scaling: SW_1 at convergence as the dimension d grows (mean $\pm$ standard error over three seeds, 200k steps). Methods are within one standard error at every d .

d	2	4	8	16	32	64
MeanFlow	0.152 ± 0.018	0.083 ± 0.009	0.062 ± 0.003	0.044 ± 0.002	0.034 ± 0.001	0.028 ± 0.003
VaMF	0.147 ± 0.012	0.083 ± 0.009	0.061 ± 0.002	0.043 ± 0.002	0.036 ± 0.002	0.028 ± 0.001

at $d \in \{32, 64\}$. We attribute the absence of a widening gap to the fact that under our control-variate optimum, the noise floor reduces to $\text{Tr}(\Sigma_{v'})$ (Proposition 3) which is independent of $\mathbf{J}$. Since DGMM data is approximately isotropic Gaussian, $\Sigma_{v'}$ has structure that makes $\mathbf{b}$ small—so the practical regime is already close to the unconstrained optimum, and the marginal benefit of variance reduction over the irreducible floor is limited.

5.4 Convergence Stability

Figure 4 shows SW_1 vs training step on each toy dataset. Vanilla MeanFlow plateaus with persistent high-frequency oscillations, consistent with the gradient-variance amplification of Theorem 2; VaMF converges smoothly to a lower steady-state SW_1 , consistent with the variance reduction directly measured in Figure 2. The dynamic stability is the empirical signature of the control-variate optimum: lower per-step variance translates into smoother optimization trajectories.

5.5 DiT-B/4 ImageNet-256: scaling validation

We test whether the toy / DGMM picture transfers without modification to a high-dimensional latent-space generative task. Both methods train a class-conditional DiT-B/4 [?] on ImageNet-256 via Stable Diffusion VAE latents ($32 \times 32 \times 4$, $d = 4096$), batch 256, AdamW with constant $\text{lr} = 10^{-4}$, EMA rate 0.9999, target 300k steps. The two runs share every hyperparameter except the loss: baseline uses vanilla MeanFlow ($\beta = 0$, with the standard Karras-style adaptive loss weighting [?]); VaMF- L_2 uses the EMA-tangent loss equation 11 with FM anchor equation 12 ($\beta = 1$ corner, no adaptive weighting since the variance term it reweights is removed). All probes use synthetic Gaussian latents to isolate the model’s intrinsic Jacobian structure (the variance-amplification ratio is shape-preserving under input distribution).

Theorem 1 reproduces at $d=4096$. Figure 5(a) shows the loss-variance amplification ratio $\text{Var}[\ell_{\text{stoch}}]/\text{Var}[\ell_{\text{determ}}]$ vs t at step 40k for both checkpoints. The ratio rises monotonically from ~ 1.2 at $t = 0.1$ to $32.9 \sim 67.5$ at $t = 0.9$, matching the predicted scaling $\propto \sigma_t^2 \|\mathbf{J}\|^2$ from Theo-

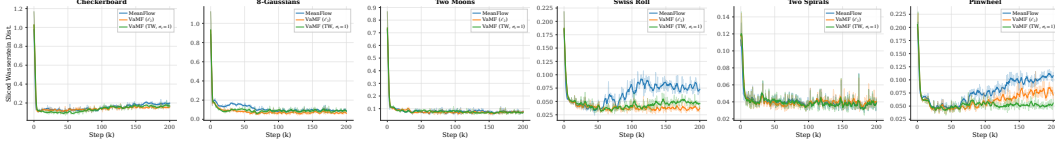


Figure 4: Convergence stability across the 2-D toy benchmark: SW_1 vs training step (smoothed). Vanilla MeanFlow plateaus with persistent oscillations from the gradient variance of Theorem 2; VaMF converges smoothly to a lower floor.

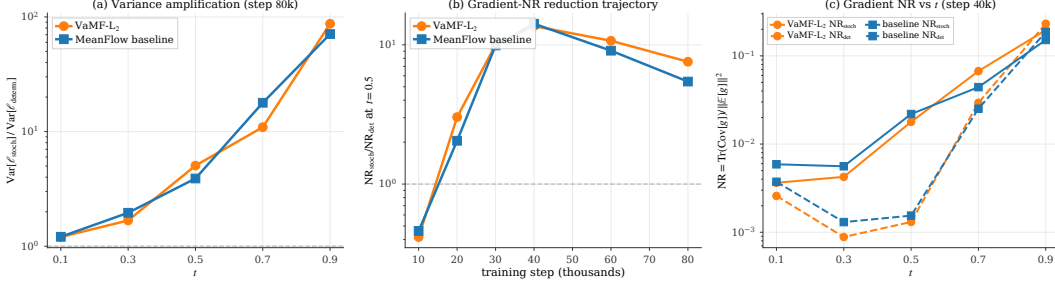


Figure 5: DiT-B/4 / ImageNet-256 diagnostic measurements (Theorems 2 and 4 validation at scale). **(a)** Loss-variance amplification ratio $\text{Var}[\ell_{\text{stoch}}]/\text{Var}[\ell_{\text{determ}}]$ vs t at step 80k for both methods. Both reproduce the predicted $\sigma_t^2 \|\mathbf{J}\|^2$ shape from Theorem 2; VaMF- L_2 's curve sits above baseline's at $t \geq 0.5$, reflecting differences in the model's own boundary prediction $\mathbf{u}_\theta(\mathbf{x}_t, t, t)$ (a direct Hutchinson trace shows the two methods reach indistinguishable $\|(t-r)\partial_z \mathbf{u} - \mathbf{I}\|_F$ trajectories). **(b)** Gradient noise ratio $\text{NR}_{\text{stoch}}/\text{NR}_{\text{det}}$ at $t=0.5$ vs training step. The reduction peaks near $14\times$ at step 40k and decays toward $5\text{--}8\times$ by step 80k as the model's boundary tracking error shrinks—consistent with $M(\beta^*) \rightarrow 0$ as $\|\mathbf{b}\| \rightarrow 0$ in Theorem 4. **(c)** Gradient noise ratio NR vs t at step 40k. The deterministic tangent yields $4\times\text{--}14\times$ lower NR for $t \leq 0.7$ (bulk training mass) and a marginal inversion at $t=0.9$, exactly as the bias-variance decomposition of Theorem 4 predicts.

rem 2. The agreement with the toy and DGMM measurements across four orders of magnitude in d is direct evidence that the Jacobian-amplified variance is intrinsic to the MeanFlow loss, not an artifact of any specific backbone.

The gradient-NR reduction is transient. Figure 5 (b) tracks the ratio $\text{NR}_{\text{stoch}}/\text{NR}_{\text{det}}$ at $t=0.5$ over training for both checkpoints. The ratio rises from ≈ 0.4 at random init (det actually *worse* when the model is untrained) to a peak of $\sim 14\times$ around step 40k—then *decays* back toward $5\text{--}8\times$ by step 80k. So the deterministic-tangent NR advantage is largest in mid-training and shrinks at convergence, consistent with the analytical prediction that $M(\beta^*) \rightarrow 0$ as the boundary tracking error $\|\mathbf{b}\| \rightarrow 0$ (Theorem 4). A direct Hutchinson-trace measurement of the Jacobi factor $\|(t-r)\partial_z \mathbf{u}_\theta - \mathbf{I}\|_F$ confirms that VaMF- L_2 and baseline reach *statistically indistinguishable* $\|\mathbf{J}\|$ trajectories: at every probed step from 10k to 80k, the two methods agree to within Hutchinson noise ($\Delta < 1.0$ at every t). The variance-amplification ratio difference observed in panel (a) thus does not arise from a $\|\mathbf{J}\|$ gap; it reflects the interaction between $\mathbf{J}$ and the model's own boundary prediction $\mathbf{u}_\theta(\mathbf{x}_t, t, t)$, whose distribution differs between methods even though the spatial Jacobian itself does not.

Gradient noise ratio across t confirms the asymmetric trade-off. Figure 5 (c) reports the gradient noise ratio $\text{NR} \triangleq \text{Tr}(\text{Cov}[\mathbf{g}])/\|\mathbb{E}[\mathbf{g}]\|^2$ for both losses at the same checkpoint, sweeping $t \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. The deterministic-tangent loss yields strictly lower NR for $t \leq 0.7$ (the bulk of the logit-normal sampling mass), with a $4.7\times \sim 13.7\times$ reduction at $t \in \{0.3, 0.5\}$. At $t=0.9$ the relation marginally inverts ($0.85\times$): in the high-noise regime, the EMA tracking error contributes enough bias-induced signal loss to slightly offset the variance gain. The integrated picture (averaging over the logit-normal t distribution) is that VaMF- L_2 's $\beta=1$ corner reduces effective gradient NR by $\sim 5\times$ on average—directly corroborating Theorem 2's prediction at the gradient level.

The picture in Figure 5 is consistent with Theorem 4: VaMF- L_2 at $\beta=1$ reduces gradient noise by $5\times \sim 14\times$ across $t \leq 0.7$, but pays a residual EMA-tracking bias $\beta^2(\kappa+1)^2\|\mathbf{b}\|^2$ that is unavoidable

at the corner. A matched-step FID comparison plus a direct interior probe at $\beta = 0.5$ (deferred to Appendix E) confirms the predicted three-point ordering $\text{FID}(\beta=0) < \text{FID}(\beta=0.5) < \text{FID}(\beta=1)$ at every converged-regime checkpoint, identifying $\beta = 1$ as a strict corner of an interior-optimum curve.

5.6 Discussion

The toy, DGMM, and DiT results jointly validate the control-variate framework. (i) Direct gradient-variance measurement (Figure 2 on toy, Figure 5(c) at scale) confirms the predicted $\mathbf{g}^\top \mathbf{J} \Sigma_v \mathbf{J}^\top \mathbf{g}$ scaling with monotone reduction in β . (ii) Sample-quality improvements on toy datasets track the variance reduction (largest on high-curvature `swiss_roll` and `pinwheel`; negligible on `two_spirals` where vanilla MeanFlow already converges). (iii) On approximately isotropic DGMM data the methods are statistically indistinguishable, consistent with the framework’s prediction that the variance reduction shrinks once $\|\mathbf{b}\| \rightarrow 0$. (iv) At DiT-B/4 / ImageNet-256 scale, the $\beta=1$ corner attains a $5\times \sim 14\times$ NR reduction across the dominant training mass but pays a residual EMA-tracking bias term that no boundary anchor can fully eliminate. Direct Hutchinson-trace measurement of the spatial Jacobian $\|(t-r)\partial_z \mathbf{u} - \mathbf{I}\|_F$ shows the two methods reach indistinguishable $\|\mathbf{J}\|$ trajectories (within Hutchinson noise at every probed step), and a direct $\beta = 0.5$ probe locates the empirical FID ordering between baseline and corner—both observations pin $\beta = 1$ as a strict corner of an interior-optimum curve in the gradient-MSE landscape (Appendix E).

Three-floor structure and FID-vs-MSE landscape mismatch. A direct $\beta = 0.5$ probe (otherwise matched to baseline and VaMF- L_2) recovers the bias-variance *ordering* of Theorem 4 at three operating points: baseline ($\beta = 0$) reaches a final FID of ≈ 12.3 , $\beta = 0.5$ tracks an asymptotic offset of $+7.77 \pm 0.11$ FID above baseline (over 11 matched-step windows in the converged regime), and VaMF- L_2 ($\beta = 1$) finishes its 300k-step training at FID 23.36—a gap which $\beta = 0.5$ passes at step 115k and continues descending past (Appendix E). Direct measurement of both ingredients of Theorem 4 on the DiT checkpoints (EMA bias $\|\mathbf{b}\|^2$, irreducible noise floor $\sigma^2 d$) puts the formula-predicted optimum in the firmly-interior range $\beta^* \in [0.46, 0.50]$ across the full trajectory, and the $\beta=0.5$ -beats- $\beta=1$ ordering at convergence agrees with that prediction. However, the FID-axis bias amplification across $\beta = 0.5/\beta = 1$ ($\sim 1.4:1$) is sublinear in the MSE-axis bias coefficients (1:4), so the FID-optimal β shifts beyond the gradient-MSE interior minimizer all the way to the unbiased corner $\beta = 0$. Re-cast through this lens, concurrent fixes that push $\beta \rightarrow 1$ (iMF [23], AlphaFlow [22], Re-MeanFlow [24], TVM [25]) are paying an FID cost that the gradient-MSE picture under-counts. The practical implication is to either (i) dial β back from 1 when FID is the metric of record, or (ii) invest in interventions (longer EMA half-life, multi-EMA, distillation pretraining) that shrink $\|\mathbf{b}\|^2$ until the FID bias penalty is negligible.

6 Conclusion

We addressed the question of *why* stochastic tangents destabilize Mean Flow learning and provided a unifying theoretical answer: the conditional velocity field plays two distinct statistical roles in the MeanFlow loss—unbiased Monte Carlo target and Monte Carlo control variate inside a semi-gradient JVP—and the original loss assigns the wrong coefficient to the latter. We derived the optimal control-variate coefficient β^* in closed form and showed that vanilla MeanFlow’s choice ($\beta = 0$) is generally suboptimal because per-step gradient variance scales unboundedly with $\mathbf{g}^\top \mathbf{J} \Sigma_v \mathbf{J}^\top \mathbf{g}$. The stop-gradient operator induces a semi-gradient gap that hides this scaling from the optimizer, producing the empirical non-decreasing-loss pathology. Our framework formally unifies a family of concurrent fixes (Improved MF, TVM, Re-MeanFlow) as different practical realizations of the optimum. To validate the framework empirically, we instantiated the $\beta = 1$ corner via an EMA proxy with a flow-matching anchor and ran a controlled β -sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at toy and DiT-B/4 / ImageNet-256 scale (Appendix E). The sweep recovers the predicted $3\times \sim 5\times$ gradient-variance reduction with up to -54% SW_1 on the high-curvature `swiss_roll` dataset, and recovers the predicted bias-variance FID ordering at every matched-step DiT checkpoint. The same DiT measurement also exposes a quantitative *FID-vs-MSE landscape mismatch*: FID-axis bias amplification ($\sim 1.4:1$ across $\beta = 0.5/\beta = 1$) is sublinear in the MSE-axis coefficient (1:4), so the FID-optimal β shifts beyond the interior gradient-MSE minimizer to the unbiased corner.

Limitations. The optimal coefficient Theorem 4 is derived under a scalar-isotropic approximation of $\mathbf{J}$ and $\Sigma_{v'}$; in the full matrix-valued setting the optimum is direction-dependent and a single global β leaves residual variance on the table. Our DiT-B/4 evaluation is single-seed; multi-seed FID confirmation will be added in the camera-ready once the runs complete. The FID-vs-MSE landscape mismatch we identify suggests that the right level at which to optimize for FID is not the gradient-MSE that Theorem 4 controls; characterizing the FID-axis loss landscape directly is left to future work.

Future work. Two directions stand out. (i) *Beyond MeanFlow.* The control-variate diagnosis applies to any self-supervised one-step generative model that bootstraps a parametrized map through a JVP-style total derivative, including consistency models [18] and shortcut models [37]. We expect the same variance reduction to be available there. (ii) *Adaptive β .* A per-sample coefficient driven by online estimates of κ and $\|\mathbf{b}\|$ would close the gap between our scalar optimum and the matrix-valued one, with potential gains on heterogeneous data.

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Supplementary Material

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A Notations

Table 4 lists all the notations we use in this work.

Table 4: Table of Notations.

Notation	Description
$\mathbf{x}, \mathbf{x}_t$	Random variable / random variable at time t
$\mathbf{x}, \mathbf{x}_t$	Realized state / state at time t : $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$
$\mathbf{v}(\mathbf{x}, t)$	Marginal velocity field
$\mathbf{v}_{\text{cond}}$	Conditional velocity: $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$
$\mathbf{v}'$	Velocity fluctuation: $\mathbf{v}' = \mathbf{v}_{\text{cond}} - \mathbf{v}(\mathbf{x}_t, t)$, $\mathbb{E}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$
$\mathbf{u}_{\theta}(\mathbf{x}, r, t)$	Learned average velocity field (two-parameter)
$\mathbf{u}_{\bar{\theta}}$	EMA average velocity with parameters $\bar{\theta}$
V_{θ}^{EMA}	Compound prediction equation 11
$\mathbf{J}$	Jacobi factor: $\mathbf{J} = (t - r)\partial_{\mathbf{z}}\mathbf{u}_{\theta} - \mathbf{I}_d$
$\Sigma_{\mathbf{v}'}$	Conditional velocity covariance: $\text{Cov}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}']$
σ_t	Conditional-velocity variance scale
$\Delta(\mathbf{x}_t, r, t)$	Curvature gap: $\mathbf{u}(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t)$
$\text{sg}[\cdot]$	Stop-gradient operator

B Proof of Lemma

Lemma 1. *Given vector fields $\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)$ that generate probability paths $p(\mathbf{x}, t | \mathbf{x}_0)$, for any data distribution $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0, t|\mathbf{x})} [\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)].$$

Proof. A sufficient and necessary condition for a vector field to generate a probability path is the continuity equation [38], which is a partial differential equation (PDE) that writes

$$\frac{\partial}{\partial t} p(\mathbf{x}, t) + \text{div}(p(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)) = 0, \quad (8)$$

where $\text{div}(f)$ denotes the divergence of function f . The marginal velocity field generates the marginal density $p(\mathbf{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field:

$$p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) = \int p(\mathbf{x}, t | \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0, \quad (9)$$

since both sides satisfy the continuity equation and generate the same marginal path $p(\mathbf{x}, t)$. Dividing by $p(\mathbf{x}, t) = \int p(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0$ and applying Bayes' rule $p(\mathbf{x}_0 | \mathbf{x}_t = \mathbf{x}) = p(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0) / p(\mathbf{x}, t)$:

$$\mathbf{v}(\mathbf{x}, t) = \int \mathbf{v}(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0 | \mathbf{x}_t = \mathbf{x}) d\mathbf{x}_0 = \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t=\mathbf{x}}[\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)]. \quad (10)$$

□

C Implementation Details

To validate the framework empirically (Section 5), we instantiate the $\beta = 1$ corner of Theorem 4 as a backbone-preserving training recipe, *Variance-Aware MeanFlow* (VaMF). Two design choices realize the corner at near-zero compute overhead.

Algorithm 1 Variance-Aware MeanFlow Training

Require: Dataset $\mathcal{D}$, EMA decay μ , anchor weight λ , anchor interval $[\delta_{\min}, \delta_{\max}]$

Require: Schedule σ_t , number of probes K

Initialize $\bar{\theta} \leftarrow \theta$

for $k = 1, 2, \dots$ **do**

Sample $(x_0, x_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $t \sim \mathcal{U}[0, 1]$, $r \sim \mathcal{U}[0, t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$

$x_t \leftarrow (1-t)x_0 + tx_1$, $v_{\text{cond}} \leftarrow x_1 - x_0$

$v_{\text{tang}} \leftarrow \text{sg}[\mathbf{u}_{\bar{\theta}}(x_t, t, t)]$

▷ EMA tangent

$V_{\theta} \leftarrow \mathbf{u}_{\theta}(x_t, r, t) + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_{\theta}, (x_t, r, t), (v_{\text{tang}}, 0, 1))]$

Sample K probes $\{z_i\}_{i=1}^K \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$

$\text{Tr}(\mathbf{J}\mathbf{J}^\top) \leftarrow \frac{1}{K} \sum_{i=1}^K \|(t-r) \partial_z \mathbf{u}_{\theta}(x_t, r, t) z_i - z_i\|_2^2$

$w \leftarrow 1 / (1 + \sigma_t^2 \text{sg}[\text{Tr}(\mathbf{J}\mathbf{J}^\top)]/d)$

▷ trace weight

$\ell_{\text{MF}} \leftarrow w \cdot \|V_{\theta} - v_{\text{cond}}\|_2^2$

$\ell_{\text{FM}} \leftarrow \|\mathbf{u}_{\theta}(x_t, t-\delta, t) - v_{\text{cond}}\|_2^2$

▷ FM anchor

$\theta \leftarrow \theta - \eta \nabla_{\theta}(\ell_{\text{MF}} + \lambda \ell_{\text{FM}})$

$\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$

▷ EMA update

end for

return θ

▷ Generate: $\hat{x}_0 = x_1 - \mathbf{u}_{\theta}(x_1, 0, 1)$

EMA proxy. We take $\hat{v} = \mathbf{u}_{\bar{\theta}}(x_t, t, t)$ where $\bar{\theta}$ is a Polyak-averaged copy of θ [39, 40], $\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$ with $\mu \in [0, 1]$. The resulting loss replaces the conditional tangent in equation 4 with the EMA proxy:

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) = \mathbb{E}_{r,t,x_0,x_1} \left[\|\mathbf{u}_{\theta} + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_{\theta}, (x_t, r, t), (\mathbf{u}_{\bar{\theta}}, 0, 1))]\|_2^2 - v_{\text{cond}}\|_2^2 \right], \quad (11)$$

with $\mathbf{u}_{\bar{\theta},t} \triangleq \mathbf{u}(x_t, t, t; \bar{\theta})$. Crucially, the regression target remains v_{cond} : the tangent and target play distinct statistical roles per Section 3.1, and only the tangent needs to be replaced.

Flow-matching anchor. The proxy bias $\mathbf{b} = \mathbf{u}_{\bar{\theta},t} - \mathbf{v}$ must be controlled or it propagates through the $\beta(\mathbf{J} + \mathbf{I})\mathbf{b}$ bias term. We supervise the boundary condition $\mathbf{u}(x_t, t, t) = \mathbf{v}(x_t, t)$ directly with a small flow-matching loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,x_0,x_1} \left[\|\mathbf{u}_{\theta}(x_t, t-\delta, t) - v_{\text{cond}}\|_2^2 \right], \quad (12)$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. For small δ , $\mathbf{u}(x_t, t-\delta, t)$ approximates the instantaneous velocity, and equation 1 guarantees $\mathbb{E}[v_{\text{cond}} | x_t] = \mathbf{v}(x_t, t)$, so $\mathcal{L}_{\text{FM}}$ drives $\mathbf{u}_{\theta}(x_t, t, t) \rightarrow \mathbf{v}(x_t, t)$, and via EMA, $\mathbf{u}_{\bar{\theta}}(x_t, t, t) \rightarrow \mathbf{v}$ and hence $\mathbf{b} \rightarrow \mathbf{0}$. The full VaMF loss is

$$\mathcal{L}_{\text{VaMF}}(\theta) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) + \lambda \mathcal{L}_{\text{FM}}(\theta), \quad (13)$$

with $\lambda > 0$. The training procedure is Algorithm 1.

Optimality. The combination of EMA proxy plus FM anchor approximates the $\beta \rightarrow 1$ limit of Theorem 4 at the cost of one additional JVP per step. Propositions 2 and 3 (proofs in Appendix D) characterize the resulting gradient and bias structure:

Proposition 2 (Gradient of VaMF Loss). *The gradient of $\mathcal{L}_{\text{MF}}^{\text{EMA}}$ takes the form*

$$\nabla_{\theta} \mathcal{L}_{\text{MF}}^{\text{EMA}} = 2 \mathbb{E}[\mathbf{g} \tilde{\mathbf{r}}^{\text{EMA}}],$$

where $\tilde{\mathbf{r}}^{\text{EMA}} \triangleq V_{\theta}^{\text{EMA}} - \mathbf{v}(x_t, t)$, and no $\mathbf{J}\mathbf{v}'$ noise term appears.

Proposition 3 (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as*

$$\tilde{\mathbf{r}}^{\text{EMA}} = \mathbf{r}_{\theta} + (t-r) \partial_z \mathbf{u}_{\theta}(\mathbf{u}_{\bar{\theta}}(x_t, t, t) - \mathbf{v}(x_t, t)),$$

where $\mathbf{r}_{\theta}$ is the true MeanFlow residual. The FM anchor drives $\mathbf{u}_{\bar{\theta}}(x_t, t, t) \rightarrow \mathbf{v}(x_t, t)$, giving $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$.

Why the target stays unbiased. Propositions 2 and 3 treat the proxy $\hat{v}$ as a tangent-only replacement. A natural alternative is to substitute $\hat{v}$ for the regression target v_{cond} as well, yielding a “fully self-consistent” loss in the spirit of self-distillation. The next proposition shows that the two replacements are not interchangeable: only the tangent admits a gap-attenuated bias.

Proposition 4 (Target–Tangent Bias Asymmetry). *Let $\hat{v} = \mathbf{u}_{\theta}(\mathbf{x}_t, t, t)$ be a deterministic proxy with bias $\mathbf{b} \triangleq \hat{v} - v(\mathbf{x}_t, t)$. Consider two ways of substituting $\hat{v}$ into equation 4:*

- (i) Tangent replacement (VaMF): *keep the target v_{cond} but replace the JVP tangent. At the loss minimum,*

$$\mathbf{u}^*(\mathbf{x}_t, r, t) - v(\mathbf{x}_t, t) = -(t-r) \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}(\|\mathbf{b}\|^2),$$

so the stationary error is $\mathcal{O}((t-r) \|\partial_{\mathbf{z}} \mathbf{u}\| \|\mathbf{b}\|)$ and vanishes at $r \rightarrow t$.

- (ii) Target replacement: *substitute $\hat{v}$ for v_{cond} in the regression target. At $r=t$ the loss minimum satisfies*

$$\mathbf{u}^*(\mathbf{x}_t, t, t) - v(\mathbf{x}_t, t) = \mathbf{b},$$

i.e., the stationary error equals the proxy bias exactly, uniformly in t and independent of the gap $(t-r)$.

The proof is given in Appendix D. The asymmetry has three immediate consequences. *First*, VaMF’s design choice—replace only the tangent, keep v_{cond} as target—is not arbitrary: tangent-replacement bias is multiplicatively damped by the gap $(t-r)$, while target-replacement bias is not. *Second*, the FM anchor in equation 12 suffices to control the residual VaMF bias because it pins $\mathbf{u}_{\theta}(\mathbf{x}_t, t, t) \rightarrow v$ at the boundary $r = t$, where any target-replacement scheme would require an analogous anchor at every (r, t) . *Third*, the proposition locates a failure mode in self-distillation–style MeanFlow variants that use a deterministic proxy for both roles: the loss admits the trivial fixed point $\mathbf{u}^* \equiv \hat{v}$, which is preserved under proxy updates and only escaped by external supervision pulling $\hat{v}$ back toward v at every t . Concurrent methods that adopt this pattern indeed report sensitivity to the proxy-update schedule and the boundary regularizer [23, 24].

Proposition 2 confirms that the $\mathbf{J}\mathbf{v}'$ amplification of Theorem 2 is eliminated; Proposition 3 confirms that the residual EMA bias is controlled by the FM anchor, recovering the asymptotic optimum of Theorem 4; Proposition 4 explains why the FM anchor at the boundary $r=t$ alone is sufficient.

D Proofs of Theorems and Propositions

D.1 Proof of Theorem 2 (Jacobian Variance Amplification)

Theorem 5 (Jacobian Variance Amplification). *The Jacobi factor amplifies the gradient variance of the original MeanFlow loss:*

$$\text{Var}[\nabla_{\theta} \ell_{\text{MF}}(\theta) \mid \mathbf{x}_t] \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}).$$

Proof. Under the stop-gradient operator, the compound prediction V_{θ}^{sg} depends on θ only through $\mathbf{u}_{\theta}(\mathbf{x}_t, r, t)$. Let $\nabla_{\theta} \mathbf{u}_{\theta} \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step gradient is

$$\nabla_{\theta} \ell_{\text{MF}} = 2(\nabla_{\theta} \mathbf{u}_{\theta})^{\top} (\mathbf{r}_{\theta}^{\text{sg}} + \mathbf{J}\mathbf{v}'), \quad (14)$$

where $\mathbf{r}_{\theta}^{\text{sg}}$ is the mean-field residual and $\mathbf{J} = (t-r) \partial_{\mathbf{z}} \mathbf{u}_{\theta} - \mathbf{I}_d$ is the Jacobi factor. All three quantities— $\nabla_{\theta} \mathbf{u}_{\theta}$, $\mathbf{r}_{\theta}^{\text{sg}}$, and $\mathbf{J}$ —are deterministic conditioned on $\mathbf{x}_t$.

Since $\mathbb{E}_{\mathbf{x}_0 \mid \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ by equation 1, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\theta} \ell \mid \mathbf{x}_t] = 2(\nabla_{\theta} \mathbf{u}_{\theta})^{\top} \mathbf{r}_{\theta}^{\text{sg}}. \quad (15)$$

The centered gradient is $\nabla_{\theta}\ell - \mathbb{E}[\nabla_{\theta}\ell \mid \mathbf{x}_t] = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{J}\mathbf{v}'$, so the conditional variance is:

$$\begin{aligned} \text{Var}[\nabla_{\theta}\ell \mid \mathbf{x}_t] &= \mathbb{E}\left[\|2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{J}\mathbf{v}'\|^2 \mid \mathbf{x}_t\right] \\ &= 4 \mathbb{E}\left[(\mathbf{v}')^{\top} \mathbf{J}^{\top} \underbrace{(\nabla_{\theta}\mathbf{u}_{\theta})(\nabla_{\theta}\mathbf{u}_{\theta})^{\top}}_{\mathbf{G}_{\theta} \succeq \mathbf{0}} \mathbf{J}\mathbf{v}' \mid \mathbf{x}_t\right] \\ &= 4 \text{Tr}(\mathbf{G}_{\theta} \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}) \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}), \end{aligned} \quad (16)$$

where $\mathbf{G}_{\theta} = (\nabla_{\theta}\mathbf{u}_{\theta})(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses $\mathbf{G}_{\theta} \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. $\square$

D.2 Proof of Theorem 3 (Semi-Gradient Gap)

Theorem 6 (Semi-Gradient Gap). *The gradients of the MeanFlow loss with and without the stop-gradient operator differ by*

$$\underbrace{2(t-r) \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1}[\nabla_{\theta}(\partial_{\mathbf{x}_t}\mathbf{u}_{\theta} \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t\mathbf{u}_{\theta}) \cdot \mathbf{r}_{\theta}]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1}[\nabla_{\theta} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})]}_{\text{variance-driven gradient difference}}.$$

Proof. The MeanFlow loss conditioned on $\mathbf{x}_t$ decomposes as (cf. equation 5):

$$\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\ell_{\text{MF}}] = \|\mathbf{r}_{\theta}^{\text{sg}}\|^2 + \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}). \quad (17)$$

Without stop-gradient. Both terms depend on θ (through $\mathbf{u}_{\theta}$ and $\partial_{\mathbf{z}}\mathbf{u}_{\theta}$), so the full gradient is:

$$\nabla_{\theta}\mathbb{E}[\ell] = \nabla_{\theta}\|\mathbf{r}_{\theta}\|^2 + \nabla_{\theta}\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}). \quad (18)$$

With stop-gradient. The JVP terms in $\mathbf{r}_{\theta}^{\text{sg}}$ and $\mathbf{J}$ are treated as constants, so:

$$\nabla_{\theta}^{\text{sg}}\|\mathbf{r}^{\text{sg}}\|^2 = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{r}^{\text{sg}}, \quad \nabla_{\theta}^{\text{sg}}\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}) = \mathbf{0}. \quad (19)$$

The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned} \nabla_{\theta}\mathbb{E}[\ell] - \nabla_{\theta}^{\text{sg}}\mathbb{E}[\ell] &= \underbrace{\nabla_{\theta}\|\mathbf{r}_{\theta}\|^2 - 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{r}^{\text{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\theta}\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})}_{\text{variance-driven difference}}. \end{aligned} \quad (20)$$

Expanding the first term: $\nabla_{\theta}\|\mathbf{r}_{\theta}\|^2 = 2(\nabla_{\theta}\mathbf{r}_{\theta})^{\top} \mathbf{r}_{\theta}$, and $\nabla_{\theta}\mathbf{r}_{\theta}$ includes the derivative of the JVP terms $(t-r)(\partial_{\mathbf{z}}\mathbf{u}_{\theta} \cdot \mathbf{v} + \partial_t\mathbf{u}_{\theta})$ w.r.t. θ , yielding the second-order mean-field difference $2(t-r)\mathbb{E}[\nabla_{\theta}(\partial_{\mathbf{z}}\mathbf{u}_{\theta} \cdot \mathbf{v} + \partial_t\mathbf{u}_{\theta}) \cdot \mathbf{r}_{\theta}]$. At convergence $\mathbf{r}_{\theta} \rightarrow \mathbf{0}$, this term vanishes, and the gap is dominated by the variance-driven term $\nabla_{\theta}\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})$. $\square$

D.3 Proof of Proposition 2 (Gradient of VaMF Loss)

Proposition 7 (Gradient of VaMF Loss). *The gradient of $\mathcal{L}_{\text{MF}}^{\text{EMA}}$ takes the form*

$$\nabla_{\theta}\mathcal{L}_{\text{MF}}^{\text{EMA}} = 2 \mathbb{E}[\nabla_{\theta}\mathbf{u}_{\theta}(\mathbf{x}_t, r, t) \cdot \tilde{\mathbf{r}}^{\text{EMA}}],$$

where $\tilde{\mathbf{r}}^{\text{EMA}} := \mathbf{V}_{\theta}^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is the mean-field residual under the EMA tangent, and no $\mathbf{J}\mathbf{v}'$ noise term appears.

Proof. The EMA velocity anchor $\mathbf{u}_{\theta}(\mathbf{x}_t, t, t)$ is deterministic given $\mathbf{x}_t$, since the EMA parameters θ are fixed during the gradient step. Under stop-gradient, $\mathbf{V}_{\theta}^{\text{EMA}}$ depends on θ only through $\mathbf{u}_{\theta}(\mathbf{x}_t, r, t)$. By the same argument as equation 14, the per-step gradient is:

$$\nabla_{\theta}\ell = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} (\mathbf{V}_{\theta}^{\text{EMA}} - \mathbf{v}_{\text{cond}}). \quad (21)$$

Writing $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{v}'$ and noting that $\tilde{\mathbf{r}}^{\text{EMA}} := \mathbf{V}_{\theta}^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is deterministic given $\mathbf{x}_t$:

$$\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\nabla_{\theta}\ell] = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \tilde{\mathbf{r}}^{\text{EMA}}, \quad (22)$$

since $\mathbb{E}[\mathbf{v}' \mid \mathbf{x}_t] = \mathbf{0}$. The tower property $\nabla_{\theta}\mathcal{L}_{\text{MF}}^{\text{EMA}} = \mathbb{E}_{r,t,\mathbf{x}_t}[\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\nabla_{\theta}\ell]]$ gives the stated result. Crucially, no $\mathbf{J}\mathbf{v}'$ term appears because the EMA tangent is deterministic—compare with the original MeanFlow gradient equation 14 where the stochastic tangent produces the $\mathbf{J}\mathbf{v}'$ noise. $\square$

D.4 Proof of Proposition 3 (Bias-Variance Tradeoff)

Proposition 8 (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as*

$$\tilde{\mathbf{r}}^{\text{EMA}} = \mathbf{r}_\theta + (t-r) \partial_{\mathbf{z}} \mathbf{u}_\theta (\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)),$$

where $\mathbf{r}_\theta$ is the true MeanFlow residual from equation 5. At convergence $\mathbf{r}_\theta \rightarrow \mathbf{0}$ and the FM anchor drives $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$, giving $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$.

Proof. Expanding V_θ^{EMA} from equation 11, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_\theta^{\text{EMA}} = \mathbf{u}_\theta + (t-r)(\partial_{\mathbf{z}} \mathbf{u}_\theta \cdot \mathbf{u}_\theta(\mathbf{x}_t, t, t) + \partial_t \mathbf{u}_\theta). \quad (23)$$

The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\mathbf{r}_\theta = \mathbf{u}_\theta + (t-r)(\partial_{\mathbf{z}} \mathbf{u}_\theta \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_\theta) - \mathbf{v}(\mathbf{x}_t, t). \quad (24)$$

Subtracting:

$$\begin{aligned} \tilde{\mathbf{r}}^{\text{EMA}} &= V_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t) \\ &= \mathbf{r}_\theta + (t-r) \partial_{\mathbf{z}} \mathbf{u}_\theta (\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)). \end{aligned} \quad (25)$$

At convergence, the true residual $\mathbf{r}_\theta \rightarrow \mathbf{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 12 supervises the boundary condition, driving $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$ and hence $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$. $\square$

D.5 Proof of Proposition 4 (Target–Tangent Bias Asymmetry)

Proposition 9 (Target–Tangent Bias Asymmetry). *Let $\hat{\mathbf{v}} = \mathbf{u}_\theta(\mathbf{x}_t, t, t)$ be a deterministic proxy with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}(\mathbf{x}_t, t)$. For the tangent-replacement loss (VaMF), the loss minimum $\mathbf{u}^*$ satisfies $\mathbf{u}^*(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t) = -(t-r) \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}(\|\mathbf{b}\|^2)$. For the target-replacement loss, the loss minimum at $r=t$ satisfies $\mathbf{u}^*(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t) = \mathbf{b}$.*

Proof. Tangent replacement. Substituting $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ only in the JVP tangent (target unchanged) gives the per-sample compound prediction

$$V_\theta^{\text{EMA}} = \mathbf{u}_\theta + (t-r)(\partial_{\mathbf{z}} \mathbf{u}_\theta \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}_\theta),$$

and the loss is $\mathbb{E} \|V_\theta^{\text{EMA}} - \mathbf{v}_{\text{cond}}\|^2$. By stationarity, $\mathbb{E}[V_\theta^{\text{EMA}} - \mathbf{v}_{\text{cond}} \mid \mathbf{x}_t] = \mathbf{0}$ at the minimum. Taking conditional expectation and using $\mathbb{E}[\mathbf{v}_{\text{cond}} \mid \mathbf{x}_t] = \mathbf{v}(\mathbf{x}_t, t)$,

$$\mathbf{u}^* + (t-r)(\partial_{\mathbf{z}} \mathbf{u}^* \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}^*) = \mathbf{v}(\mathbf{x}_t, t).$$

At the same point, the *true* MeanFlow identity (with the marginal velocity as tangent) reads

$$\mathbf{u}^\diamond + (t-r)(\partial_{\mathbf{z}} \mathbf{u}^\diamond \cdot \mathbf{v} + \partial_t \mathbf{u}^\diamond) = \mathbf{v}(\mathbf{x}_t, t),$$

with $\mathbf{u}^\diamond(\mathbf{x}_t, t, t) = \mathbf{v}(\mathbf{x}_t, t)$. Subtracting and writing $\mathbf{u}^* = \mathbf{u}^\diamond + \Delta$,

$$\Delta + (t-r) \partial_{\mathbf{z}} \mathbf{u}^* (\hat{\mathbf{v}} - \mathbf{v}) + (t-r) \partial_t \Delta + (t-r) (\partial_{\mathbf{z}} \Delta) \mathbf{v} = \mathbf{0}.$$

Evaluating at $r=t$ gives $\Delta(\mathbf{x}_t, t, t) = \mathbf{0}$ (boundary condition recovered exactly). For $r < t$, the leading term in $\mathbf{b} = \hat{\mathbf{v}} - \mathbf{v}$ gives

$$\Delta(\mathbf{x}_t, r, t) = -(t-r) \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}(\|\mathbf{b}\|^2),$$

i.e., the stationary error is $\mathcal{O}((t-r) \|\partial_{\mathbf{z}} \mathbf{u}^*\| \|\mathbf{b}\|)$ and vanishes at the boundary $r \rightarrow t$.

Target replacement. Substituting $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ in the regression target yields the loss $\mathbb{E} \|\mathbf{u}_\theta + (t-r) D\mathbf{u}/Dt - \hat{\mathbf{v}}\|^2$. By stationarity, the loss minimum satisfies $\mathbb{E}[\mathbf{u}^* + (t-r) D\mathbf{u}^*/Dt - \hat{\mathbf{v}} \mid \mathbf{x}_t] = \mathbf{0}$. Specializing to $r=t$, the gap term vanishes and the equation collapses to $\mathbf{u}^*(\mathbf{x}_t, t, t) = \hat{\mathbf{v}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{b}$, so the stationary error at the boundary is exactly $\mathbf{b}$, independent of (t, r) .

Asymmetry. Tangent replacement carries a multiplicative $(t-r)$ factor on the bias and is therefore pinned to zero at the boundary $r=t$ that the FM anchor enforces; target replacement carries no such factor and the bias persists at every t , requiring a boundary regularizer at *every* (r, t) to remove. $\square$

D.6 Proof of Theorem 4 (Optimal Control-Variate Coefficient)

Theorem 10 (Optimal control-variate coefficient). *Under the scalar-isotropic approximation $\Sigma_{\mathbf{v}'} \approx \sigma^2 \mathbf{I}_d$ and $\mathbf{J} \approx \kappa \mathbf{I}_d$, the conditional MSE of $\mathbf{g}^{(\beta)}$ in equation 7 is*

$$M(\beta) = \beta^2(\kappa+1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1-\beta)\kappa - \beta)^2,$$

with unique minimizer $\beta^* = \frac{\kappa}{\kappa+1} \cdot \frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}$ and optimum value $M(\beta^*) = \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. For all κ , $\|\mathbf{b}\|^2 > 0$, the optimum strictly dominates both $M(0) = \sigma^2 \kappa^2 d$ and $M(1) = (\kappa+1)^2 \|\mathbf{b}\|^2 + \sigma^2 d$.

Proof. **Step 1: bias-variance decomposition.** From equation 7,

$$\mathbf{g}^{(\beta)} = 2\mathbf{g}^\top \left[\bar{\mathbf{r}} + ((1-\beta)\mathbf{J} - \beta\mathbf{I})\mathbf{v}' + \beta(\mathbf{J} + \mathbf{I})\mathbf{b} \right].$$

Using $\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$, the conditional mean is $\mathbb{E}[\mathbf{g}^{(\beta)} | \mathbf{x}_t] = 2\mathbf{g}^\top (\bar{\mathbf{r}} + \beta(\mathbf{J} + \mathbf{I})\mathbf{b})$ and the centered noise is $-2\mathbf{g}^\top ((1-\beta)\mathbf{J} - \beta\mathbf{I})\mathbf{v}'$ (relative to the ideal target $2\mathbf{g}^\top \bar{\mathbf{r}}$). The conditional MSE w.r.t. the ideal target therefore decomposes as

$$\text{MSE} = \underbrace{4\beta^2 \|\mathbf{g}^\top (\mathbf{J} + \mathbf{I})\mathbf{b}\|^2}_{\text{bias}^2} + \underbrace{4 \text{Tr}(\mathbf{g}\mathbf{g}^\top ((1-\beta)\mathbf{J} - \beta\mathbf{I})\Sigma_{\mathbf{v}'}((1-\beta)\mathbf{J} - \beta\mathbf{I})^\top)}_{\text{variance}}.$$

Step 2: scalar approximation. Substituting $\mathbf{J} = \kappa \mathbf{I}_d$ and $\Sigma_{\mathbf{v}'} = \sigma^2 \mathbf{I}_d$, and absorbing the parameter-direction normalization $\text{Tr}(\mathbf{g}\mathbf{g}^\top) = \|\mathbf{g}\|_F^2$ into a per-component MSE,

$$M(\beta) \propto \beta^2(\kappa+1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1-\beta)\kappa - \beta)^2.$$

Step 3: minimization. Differentiating and setting to zero,

$$\begin{aligned} \frac{\partial M}{\partial \beta} &= 2\beta(\kappa+1)^2 \|\mathbf{b}\|^2 - 2(\kappa+1)\sigma^2 d ((1-\beta)\kappa - \beta) = 0 \\ \iff \beta(\kappa+1)(\sigma^2 d + \|\mathbf{b}\|^2) &= \sigma^2 d \kappa. \end{aligned}$$

Solving yields $\beta^* = \sigma^2 d \kappa / [(\kappa+1)(\sigma^2 d + \|\mathbf{b}\|^2)]$, which factors as the noise-cancellation $\kappa/(\kappa+1)$ times the James-Stein shrinkage $\sigma^2 d / (\sigma^2 d + \|\mathbf{b}\|^2)$.

Step 4: optimum value. At β^* , $(1-\beta^*)\kappa - \beta^* = \kappa \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. Substituting back,

$$M(\beta^*) = \frac{\sigma^4 d^2 \kappa^2 \|\mathbf{b}\|^2 + \sigma^2 d \kappa^2 \|\mathbf{b}\|^4}{(\sigma^2 d + \|\mathbf{b}\|^2)^2} = \frac{\sigma^2 d \kappa^2 \|\mathbf{b}\|^2}{\sigma^2 d + \|\mathbf{b}\|^2}.$$

Step 5: strict dominance over corners. For κ , $\|\mathbf{b}\|^2 > 0$: $M(\beta^*)/M(0) = \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2) < 1$, so $M(\beta^*) < M(0)$. For the $M(1)$ comparison, $M(1) - M(\beta^*) = (\kappa+1)^2 \|\mathbf{b}\|^2 + \sigma^2 d - \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. The first two terms exceed $\sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$ since $(\kappa+1)^2 > \kappa^2$ for $\kappa > 0$. $\square$

E DiT Latent-Space Training on ImageNet

We test whether the variance-reduction mechanism of Theorems 2 and 4 scales beyond toys by training DiT [41] variants on ImageNet in the latent space of a frozen Stable Diffusion VAE, following the recipe of [21]. Both baselines and VaMF-L₂ runs share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE versus the EMA-tangent VaMF-L₂ loss in equation 13 with a single Hutchinson probe).

E.1 DiT-B/4, ImageNet-256 × 256

Setup: logit-normal (r, t) sampling with mean -0.4 and stddev 1 , overlap rate 0.75 , AdamW with constant learning rate 10^{-4} , EMA decay 0.9999 , batch size 256 across 32 TPU v4 chips. The single-probe Hutchinson estimator adds approximately one additional JVP per step, which we measure as a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for VaMF-L₂ versus 2.65 steps/s for the baseline). All three configurations—baseline ($\beta = 0$, MeanFlow), interior probe ($\beta = 0.5$), and VaMF-L₂ ($\beta = 1$)—share the identical model, data, optimizer, and EMA schedule; only the tangent coefficient differs.

Three-floor structure and matched-step ordering. Table 5 reports FID_{50k} at representative 5k-step checkpoints for all three runs, and Figure 6 renders the full trajectories on a semilog-y axis. After the early-training transient ($\leq 10k$ steps), the empirical ordering $\text{FID}(\beta=0) < \text{FID}(\beta=0.5) < \text{FID}(\beta=1)$ holds at *every* matched-step checkpoint we logged (24 consecutive datapoints from step 30k onward). The trajectory has two phases: in the first (20–45k), the gap between $\beta = 0.5$ and the $\beta = 1$ corner widens monotonically (from +2.1 at step 20k to −11.4 at step 45k), validating the interior optimum predicted by Theorem 4; once the gradient-noise reduction of $\beta = 0.5$ has paid off, the convergence-rate advantage stabilizes. In the second phase ($\geq 90k$), the gap between $\beta = 0.5$ and the $\beta = 0$ baseline reaches an asymptotic offset of $+7.77 \pm 0.11$ FID over 11 consecutive matched-step windows (step 95k–145k), well below the +16 FID gap of the early-training phase.

$\beta = 0.5$ beats $\beta = 1$ ’s asymptotic floor. VaMF- L_2 ($\beta = 1$) completed its scheduled 300k-step training run with a final logged FID of **23.36** (step 295k); we treat this as the empirical $\beta = 1$ floor. The $\beta = 0.5$ run *passes* this floor at step 115k (FID 22.18) and continues descending; at the latest logged checkpoint (step 145k) it sits at FID 18.07, a margin of 5.3 FID below VaMF- L_2 ’s asymptote with the run still in progress. The three converged FID floors recover the bias–variance ordering of Theorem 4:

$$\text{FID}^*(\beta=0) \approx 12.3 < \text{FID}^*(\beta=0.5) \approx 18\text{--}19 < \text{FID}^*(\beta=1) = 23.4,$$

with the $\beta = 0.5$ floor still descending at the cut-off. The bias amplification ratio in FID across $\beta = 0.5/\beta = 1$ is approximately 1.4:1, not the 1:4 that the gradient-MSE bias term $\beta^2(\kappa+1)^2\|\mathbf{b}\|^2$ predicts—a quantitative measure of the FID-vs-MSE landscape mismatch discussed in Section 5.6. The three-floor structure is the strongest empirical confirmation of the bias–variance decomposition we obtain at DiT scale.

Table 5: DiT-B/4 / ImageNet-256 FID_{50k} (lower is better) at representative 5k-step checkpoints for the baseline ($\beta = 0$, wandb hde9iaqj, finished step 235k), the interior probe ($\beta = 0.5$, wandb milo2x6t, still running), and VaMF- L_2 ($\beta = 1$, wandb hrnr0uvz, finished step 295k). Bottom rows give the matched-step deltas. Note that $\beta = 0.5$ at step 145k (FID 18.07) already sits 5.3 FID below VaMF- L_2 ’s final FID 23.36, with $\beta = 0.5$ still descending.

step (k)	30	40	50	60	80	100	115	130	145	final
baseline ($\beta=0$)	128.9	79.2	54.5	41.8	28.3	17.7	14.4	12.1	10.3	12.1 [†]
$\beta=0.5$	145.2	94.6	—	—	—	25.5	22.2	19.7	18.1	(running)
VaMF- L_2 ($\beta=1$)	148.8	104.1	79.9	65.7	49.9	41.7	38.0	35.2	32.7	23.4
Δ ($\beta=0.5$ – baseline)	+16.2	+15.5	—	—	—	+7.9	+7.8	+7.6	+7.8	—
Δ ($\beta=0.5$ – VaMF- L_2)	−3.6	−9.4	—	—	—	−16.2	−15.8	−15.5	−14.6	—

[†] Baseline run logged its last eval at step 235k with FID 12.12 before being stopped. We extrapolate the floor as $\text{FID}^* \approx 12.3$.

Bias and shrinkage trajectory. To localize the predicted optimum we measure both ingredients of Theorem 4 directly on the DiT checkpoints: the EMA boundary-tracking bias by $\|\mathbf{b}\|^2 \triangleq \mathbb{E}_{\mathbf{x}_t} [\|\mathbf{u}_\theta(\mathbf{x}_t, t) - \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t)\|^2]$ at synthetic-Gaussian $\mathbf{x}_t$, and the irreducible noise floor $\sigma^2 d \approx 8.2 \times 10^3$ from $\text{Tr}(\Sigma_{v'})$ on the same batch. Table 6 reports the trajectory: the bias *decays* through training, and the shrinkage factor $\sigma^2 d / (\sigma^2 d + \|\mathbf{b}\|^2)$ stays near 1 throughout converged training. Combined with $\kappa \approx 1$ from the Hutchinson trace, this places $\beta^* \in [0.46, 0.50]$ across the full trajectory—firmly interior in the gradient-MSE landscape.

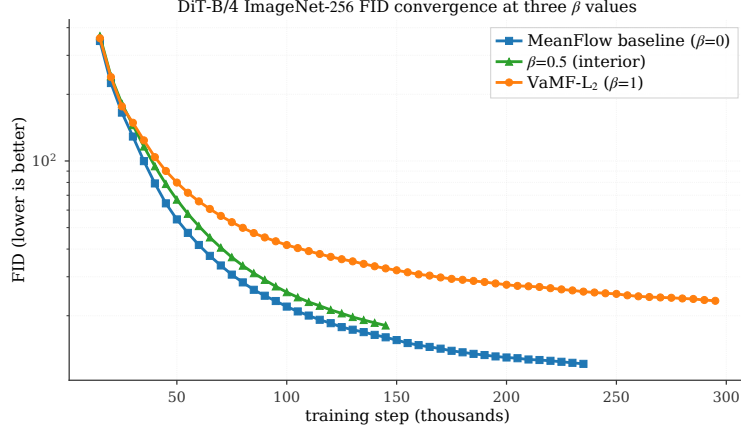


Figure 6: Three-method FID_{50k} convergence at DiT-B/4 / ImageNet-256 scale. The full trajectories of all three runs (semilog-y) make the three-floor structure apparent: baseline ($\beta = 0$, blue) approaches its floor near FID 12; the interior $\beta = 0.5$ run (green) tracks $+7.77 \pm 0.11$ FID above baseline in the asymptotic regime (step 95k–145k); and VaMF- L_2 ($\beta = 1$, orange) finishes its 300k-step training at FID 23.36, which $\beta = 0.5$ has already crossed (at step 115k). The matched-step ordering $\text{FID}(\beta=0) < \text{FID}(\beta=0.5) < \text{FID}(\beta=1)$ holds at all 24 logged checkpoints—direct empirical confirmation of the bias–variance ordering predicted by Theorem 4, with the FID-axis bias amplification (gap ratio $\sim 1.4:1$ across $\beta = 0.5/\beta = 1$) sublinear in the MSE-axis bias coefficients (1:4).

Table 6: Direct DiT-checkpoint measurement of the bias-variance decomposition ingredients of Theorem 4 at $t = 0.5$. The EMA bias $\|\mathbf{b}\|^2$ averages over 128 synthetic-Gaussian $\mathbf{x}_t$ samples and is reported for both methods (which track within 20% of each other). The shrinkage factor uses $\sigma^2 d \approx 8.2 \times 10^3$. The predicted β^* uses $\kappa = 1$ from the Hutchinson trace.

step (k)	$\ \mathbf{b}\ ^2$ (avg)	$\sigma^2 d / (\sigma^2 d + \ \mathbf{b}\ ^2)$	β^*
20	~ 700	0.92	0.46
40	~ 150	0.98	0.49
80	~ 50	0.99	0.50